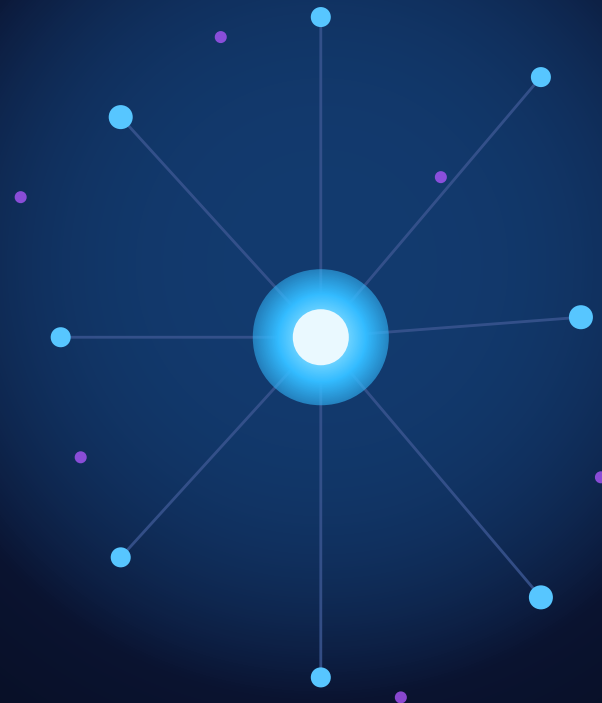


Advanced **GenAI** **Engineer** Program

A live, mentor-led program for engineers who want to design, evaluate, deploy and defend production-grade GenAI systems — not just build demos.

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| 16 Weeks | Live Online | 8-10 PM IST | 20 Engineers Max | ₹49,999 one-time



ABOUT COGNIVARA

Learn technology the way it's actually built.

Cognivara Academy is a live, mentor-led learning academy focused on Generative AI, Data Science, Cloud and DevOps. We believe engineers learn best by building, breaking, debugging and shipping — not by watching a syllabus go by.

THE PROMISE

Stop building demos. Start building production systems.

You'll learn by building: a deployed GenAI API, an evaluated retrieval system, a fine-tuned model, and the architecture judgment to defend your design choices in a technical interview.

RAG

Hands-on vector databases, hybrid search and evaluation.

Agents

Built with LangGraph, MCP, guardrails and security.

Fine-tuning

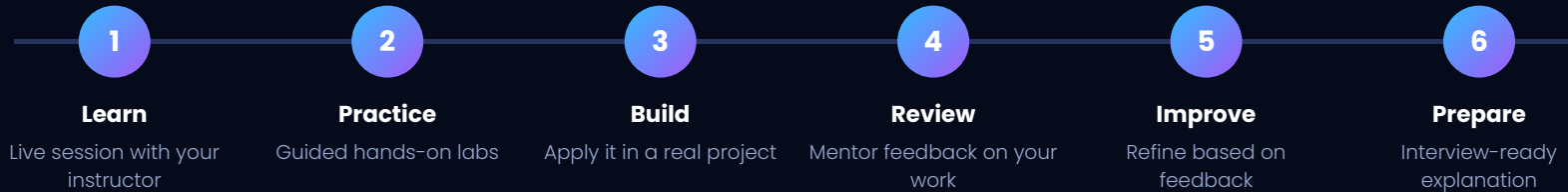
LoRA, QLoRA, distillation and real serving workflows.

Production

Docker, AWS, CI/CD and cost-aware deployment.

WHY LEARN WITH US

Built around a simple learning cycle



Live instructor-led

Interactive online sessions, not pre-recorded lectures.

Career support

Resume, GitHub, mock interviews, job-search strategy.

Session recordings

Every live session is recorded for enrolled learners.

Small cohorts

Capped at 20 engineers for real mentor attention.

1:1 doubt support

Dedicated time to clear technical blockers.

Current curriculum

Tools and workflows reviewed as the industry changes.

Project reviews

Direct feedback on your implementation.

Portfolio-ready

A capstone you can show in interviews and on GitHub.

WHO IT'S FOR

Built for engineers ready to go deeper

Designed for

- Software engineers and backend / full-stack developers
- ML engineers and data scientists
- Professionals transitioning into GenAI roles
- Students and professionals with an IT background

Not the right fit for

- Absolute beginners without a coding background
- Learners who want to avoid hands-on debugging
- People looking for a passive, video-only format
- Anyone expecting a guaranteed job outcome

Prerequisites

- Comfort with Python, basic SQL and Git/GitHub
- Familiarity with REST APIs and foundational ML concepts
- A laptop with a reliable internet connection

Prior GenAI experience helps but is not required.

PROGRAM PROGRESSION

Six phases. One evolving system.

Each phase builds on the last. You ship real, working components incrementally instead of only learning concepts in isolation — by week 16 they come together into one capstone system.

Weeks 1–3

Engineering Foundations

FastAPI, async Python, embeddings, chunking and transformer architecture.

Weeks 4–7

Production RAG

Vector search, hybrid retrieval, reranking and automated RAG evaluation.

Weeks 8–10

Agents, Tools & Security

LangGraph, tool calling, MCP, prompt-injection defense and guardrails.

Weeks 11–13

Fine-Tuning Engineering

LoRA, QLoRA, evaluation against overfitting, distillation and serving.

Week 14

Deploy

Docker, CI/CD, AWS EC2/ECR, caching, cost tracking and model routing.

Weeks 15–16

Integrate & Defend

Full system assembly, live demo, architecture defense and interview prep.

PHASE 1 · WEEKS 1-3

Engineering Foundations

Week 1 — FastAPI & Async Python

Async/await, Pydantic, streaming responses, server-sent events and non-blocking LLM calls.

Week 2 — Embeddings & Chunking

Tokenization, semantic chunking, local LLMs and INT8/INT4 quantization.

Week 3 — Transformer Architecture

Self-attention, KV cache, time-to-first-token, speculative decoding and inference benchmarking.

PHASE 2 · WEEKS 4-7

Production RAG Systems

Build retrieval systems that are accurate, fast and measurable — not just functional.

- Vector search: FAISS to Qdrant, HNSW indexing, metadata filtering
- Hybrid search (BM25 + dense retrieval) with cross-encoder reranking
- Advanced strategies: RAG-Fusion, HyDE, multi-query, reciprocal rank fusion
- Evaluation and observability: RAGAS, LLM-as-judge, LangSmith, Phoenix tracing

PHASE 3 · WEEKS 8-10

Agents, Tools & Security

Week 8 — LangGraph & Stateful Agents

Graph-based workflows, conditional routing, cyclic graphs, memory and planning.

Week 9 — Tool Calling & MCP

Function schemas, MCP architecture, multi-agent coordination and human-in-the-loop review.

Week 10 — Security & Guardrails

Prompt-injection defense, input/output validation and least-privilege tool access.

PHASE 4 · WEEKS 11-13

Fine-Tuning Engineering

If you need	Use this approach
Current information or citations	RAG
A consistent output format	Prompting
A specific style or behaviour	Fine-tuning
Lower inference cost at scale	Fine-tuning
External actions	Agents / tools

Weeks 11-13 cover instruction tuning, LoRA/QLoRA, dataset construction, evaluating against overfitting and catastrophic forgetting, and serving via distillation and quantization.

PHASE 5-6 · WEEKS 14-16

Deploy, Integrate & Defend

The final three weeks bring every earlier phase together into one deployed, defensible system.

Week 14 — Deploy

Docker, CI/CD, AWS EC2/ECR, Redis caching, cost tracking and model routing.

Week 15 — Integrate

Full system assembly, end-to-end testing, architecture review, cost and performance analysis.

Week 16 — Defend

Live demo, architecture defense, interview preparation and a portfolio-ready deliverable.

Capstone principle

You deploy a fine-tuned model only if evaluation proves it is the better engineering choice over RAG or prompting — not by default.

CAPSTONE PROJECT

A real production system you'll deploy

Build one end-to-end GenAI system across a domain of your choice — Finance, Legal, Healthcare or Enterprise — designed to ship, measured with evidence, and strong enough for your portfolio.

- A fully deployed, production-ready GenAI application
- An evaluation report with metrics and trade-offs
- Cost, latency and quality analysis
- Architecture documentation you can present confidently
- A public GitHub project recruiters can review

CAPSTONE RUBRIC

How the capstone is evaluated

Component	Weight
RAG quality & evaluation	20%
System architecture	15%
Agent / tool reliability	15%
Fine-tuning methodology	15%
Security	10%
Performance & cost	10%
Deployment	10%
Technical presentation	5%

TOOLS & TECHNOLOGIES

What you'll work with

Python

FastAPI

LLM APIs

Embeddings

FAISS

Qdrant

RAG

LangGraph

MCP

RAGAS

LangSmith

LoRA

QLoRA

Docker

AWS

CI/CD

Redis

GitHub Actions

INSTRUCTOR

Learn from someone who has shipped it

Roshan Salunke

Data Science Engineer

- 2+ years building production AI/ML and GenAI applications
- Expertise in LLMs, agentic AI, RAG pipelines and scalable data systems
- Built a financial RAG assistant serving 1,660+ companies and 73,000+ reports
- Reduced inference latency from 3.0s to 1.2s and storage by 94%
- Trained 100+ engineers in data science and NLP with 90%+ satisfaction

SCHEDULE & FEE

16 weeks. About 160 live hours. One fee.

16

WEEKS

10

HOURS / WEEK

160

TOTAL LIVE HOURS

20

MAX COHORT SIZE

Program fee includes

- Live sessions with recordings
- Hands-on labs and projects
- 1:1 doubt-solving support
- Project reviews and feedback
- Career and interview preparation
- Certificate on completion

Program format

Fee	₹49,999 one-time
EMI	Not available
Time slot	8:00–10:00 PM IST, Mon–Fri
Cloud / GPU budget	₹2,000–5,000 (external, as used)
Certificate	Yes, on completion requirements
Placement guarantee	No

We provide career and placement assistance — resume review, GitHub/portfolio review, mock interviews and job-search strategy. This does not constitute a guarantee of employment, salary, company or role.

FREQUENTLY ASKED QUESTIONS

Got questions? We've got answers.

Do I need prior GenAI experience?

No, but Python proficiency is required to keep pace with the labs.

What if I miss a live session?

Recordings are available for enrolled learners for every live session.

Do I need to pay for GPU or cloud usage?

Budget roughly ₹2,000–5,000 for GPU/AWS usage during the program.

Is placement guaranteed?

No. We provide placement assistance, not a guarantee of employment.

Will I fine-tune a real model?

Yes — you'll apply QLoRA to a small open-weight model during Phase 4.

Is there a certificate?

Yes, awarded on meeting the program's completion requirements.

Are EMI plans available?

Not currently — the program fee is a one-time payment.

Can I use the capstone in my portfolio?

Yes, it's designed as a public GitHub project you can show recruiters.

ENROLLMENT PROCESS



Ready to build your next skill?

Learn with mentors. Build real projects. Prepare for your next opportunity.

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Placement assistance does not constitute a job guarantee. Program fees, schedules, curriculum and access terms are subject to the applicable enrollment terms shared before payment.